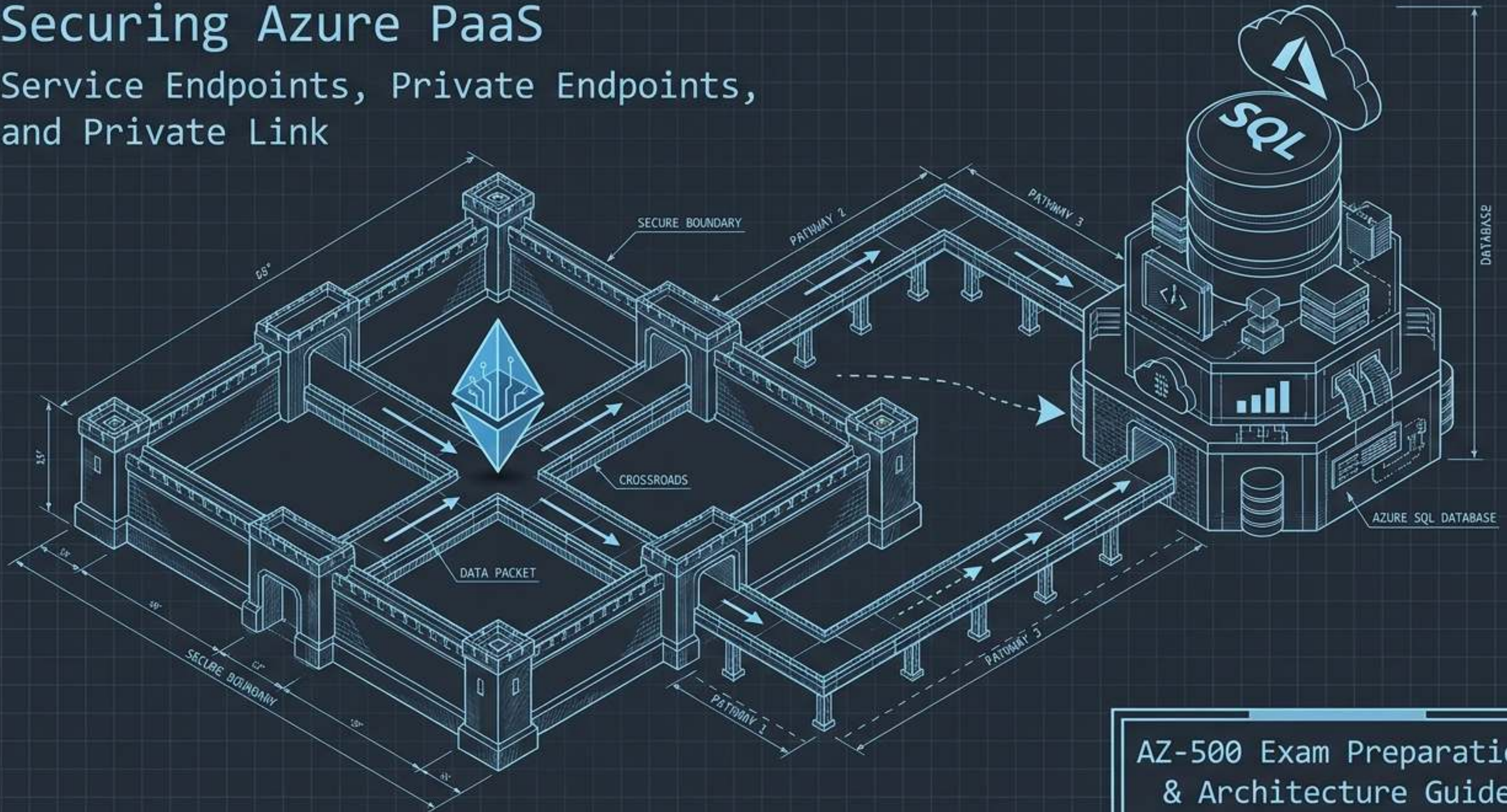


Securing Azure PaaS

Service Endpoints, Private Endpoints,
and Private Link



AZ-500 Exam Preparation
& Architecture Guide

THE DEFAULT POSTURE IS PUBLIC BY DEFAULT

NOTE 1.0: BASELINE CONFIGURATION

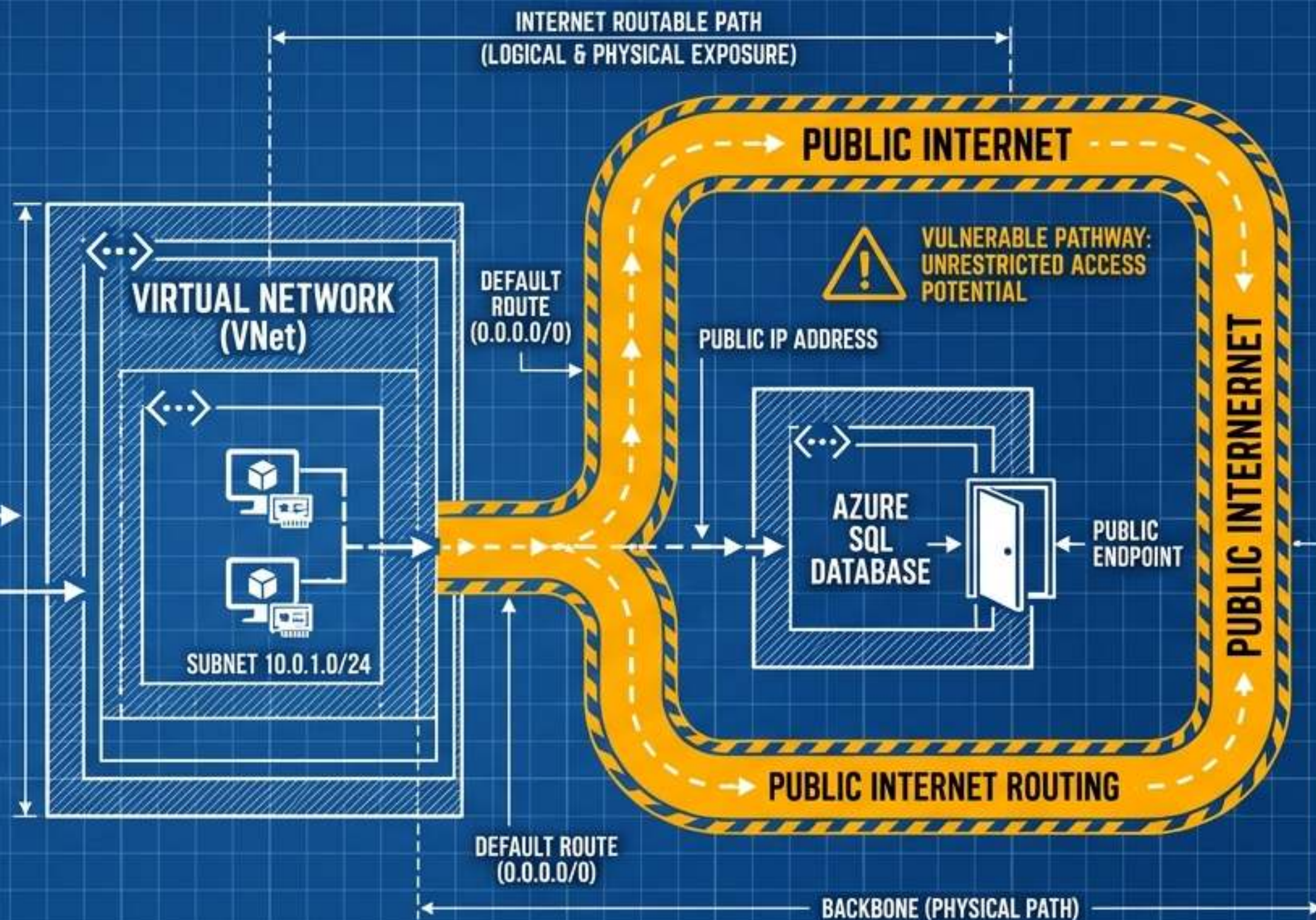
The Baseline: Azure PaaS resources (Storage, SQL, Key Vault) are provisioned with public, internet-resolvable IPs.

NOTE 2.0: ROUTING MECHANICS

The Routing Reality: Default traffic from a VNet to PaaS uses public endpoints.

NOTE 3.0: SECURITY MISCONCEPTION

The Illusion of Safety: Traffic may stay physically on the Microsoft backbone, but it remains logically internet-facing. Backbone routing does not equal network isolation.



THE DEFAULT POSTURE IS PUBLIC BY DEFAULT

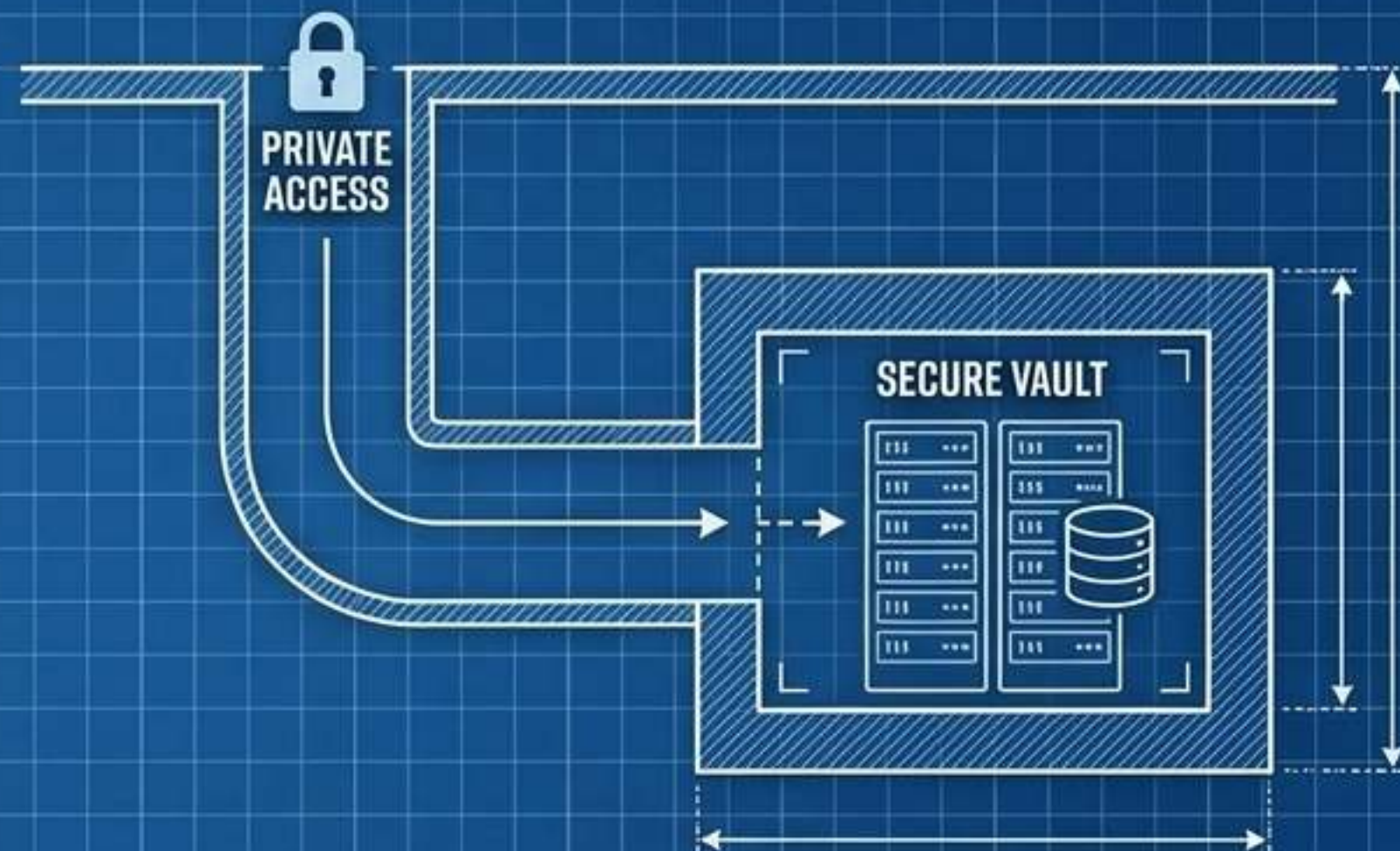
SCALE		
SCALE		

AUTHENTICATION $\neq$ NETWORK SECURITY



WHO HAS THE KEY

Identity Controls (RBAC/Tokens): Determine who is allowed in. If the door is on a public street, anyone worldwide can attempt to guess the key.



WHERE THE DOOR IS LOCATED

Network Controls (Endpoints): Determine where the door is located. Our goal is to move the door off the public street to eliminate external access attempts entirely.

SERVICE ENDPOINTS: EXTENDING VNET IDENTITY

THE MECHANISM:

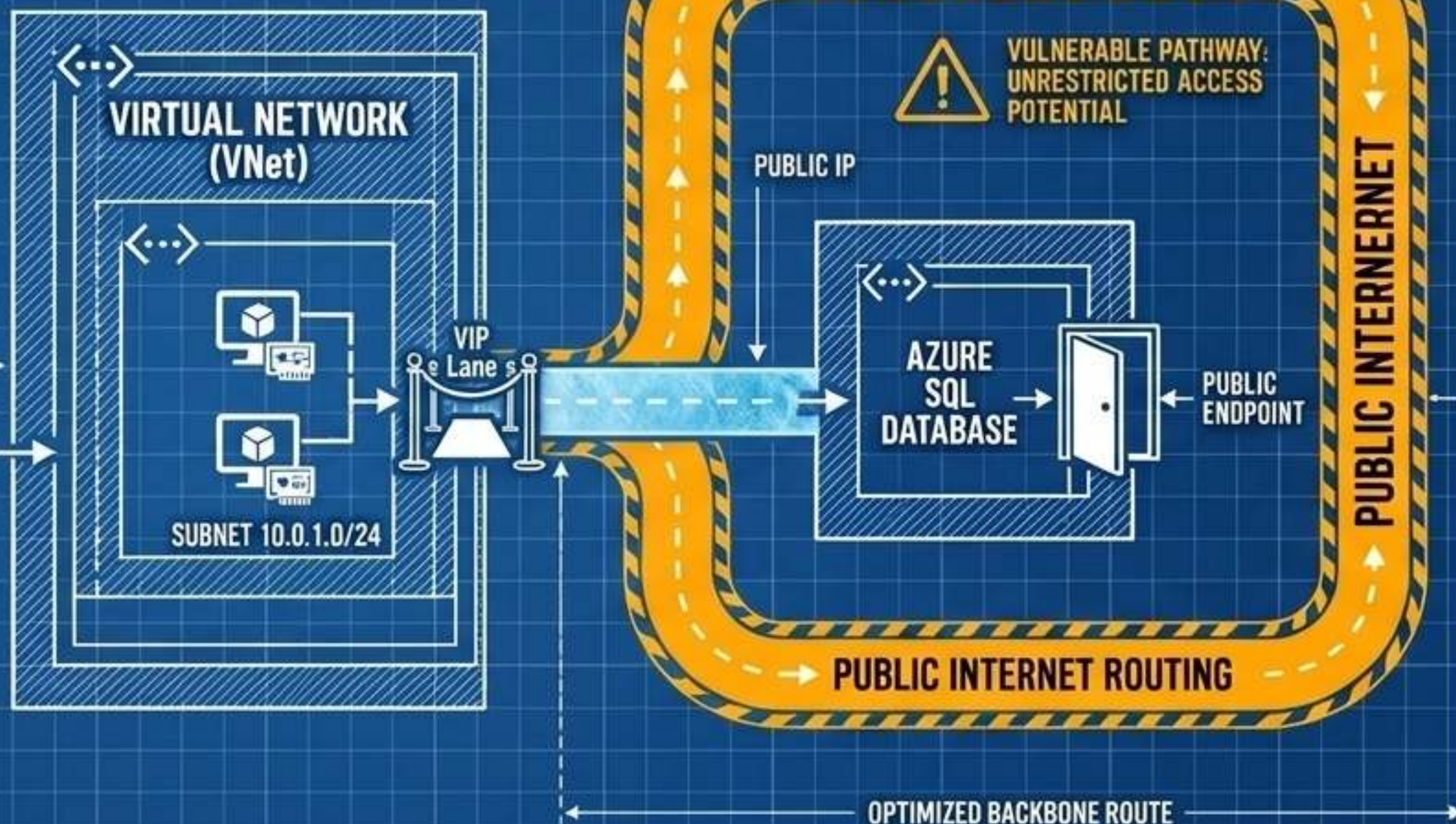
PaaS firewalls are configured to recognize and accept traffic directly from your specific VNet's private IP space.

THE CONFIGURATION:

Applied at the Subnet level.

THE BENEFIT:

Optimized routing exclusively over the Azure Backbone. Low friction implementation.



TITLE

**SERVICE ENDPOINTS:
EXTENDING VNET IDENTITY**

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THE SERVICE ENDPOINT CATCH

THE DOOR REMAINS PUBLIC:

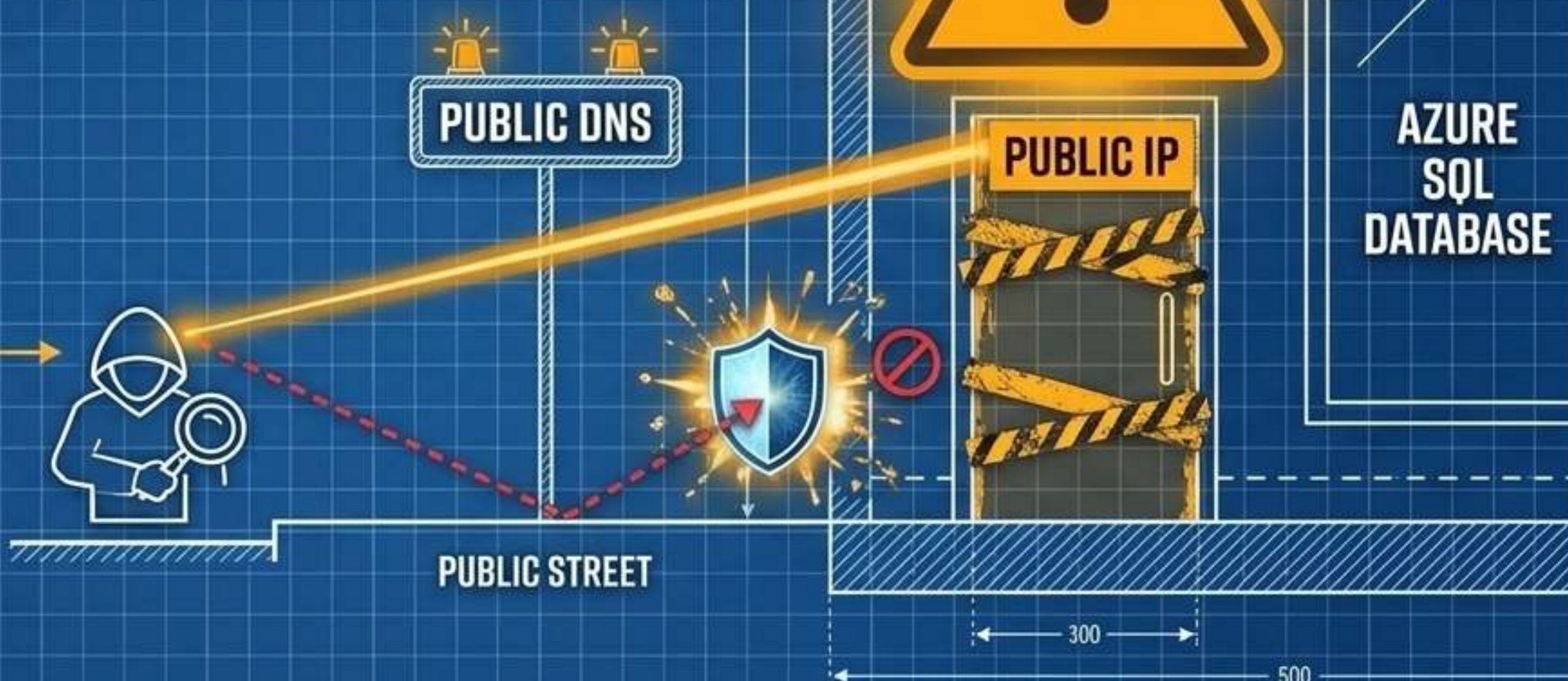
The PaaS endpoint is still internet-reachable. The firewall merely drops unauthorized traffic.

PUBLIC DNS:

The FQDN still resolves to a Public IP address.

EXFILTRATION RISK:

Because it applies to the service (e.g., all Azure Storage) rather than a specific instance, it does not prevent data exfiltration to attacker-controlled tenants.



THE SERVICE ENDPOINT CATCH: REMAINING VULNERABILITIES

PROJECTED

AZURE SECURITY ARCHITECTURE

SAFE

BSSER

PRIVATE ENDPOINTS: TRUE NETWORK ISOLATION

THE MECHANISM:

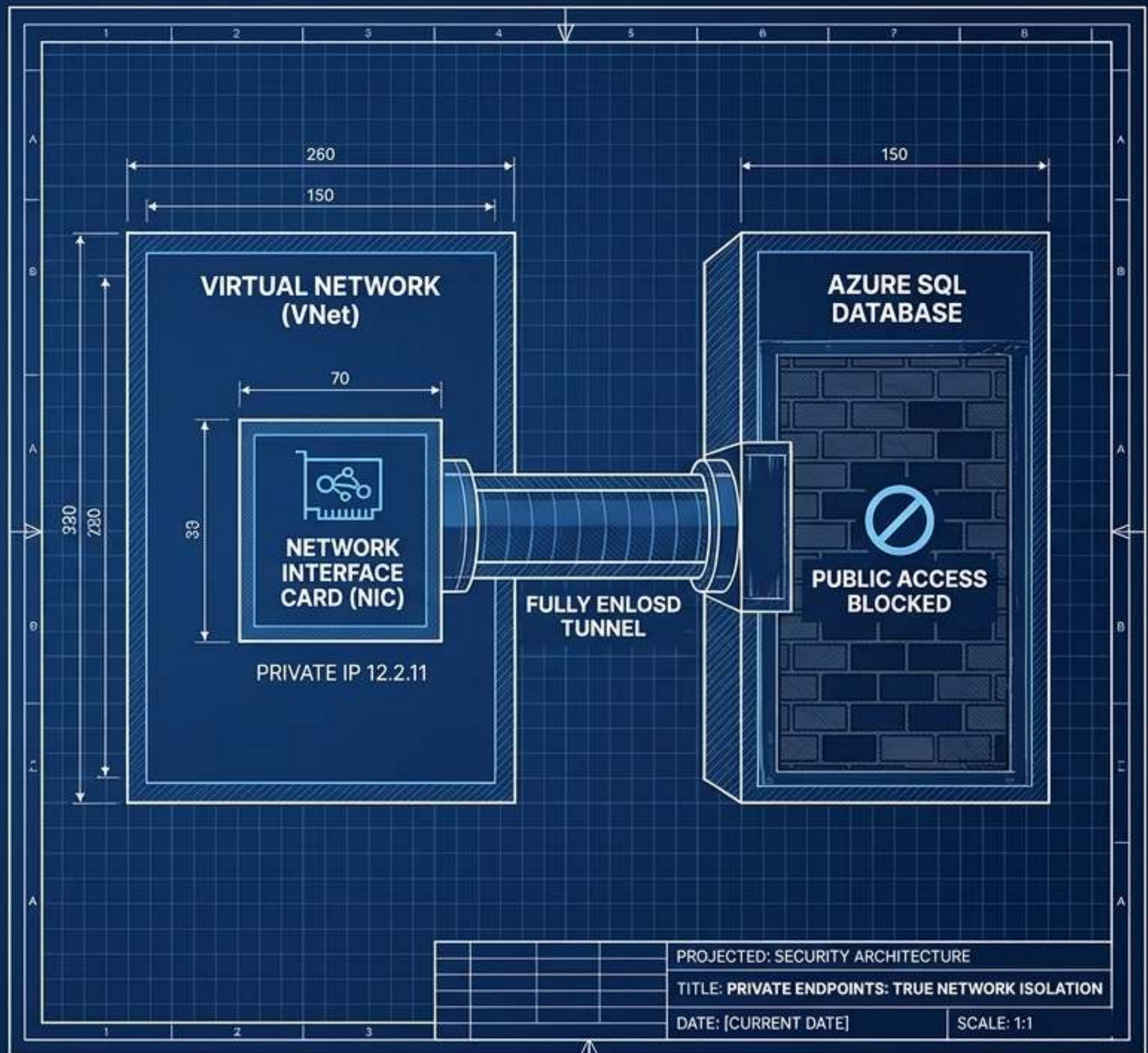
Deploys a dedicated NIC with a private IP directly inside your VNet.

THE SCOPE:

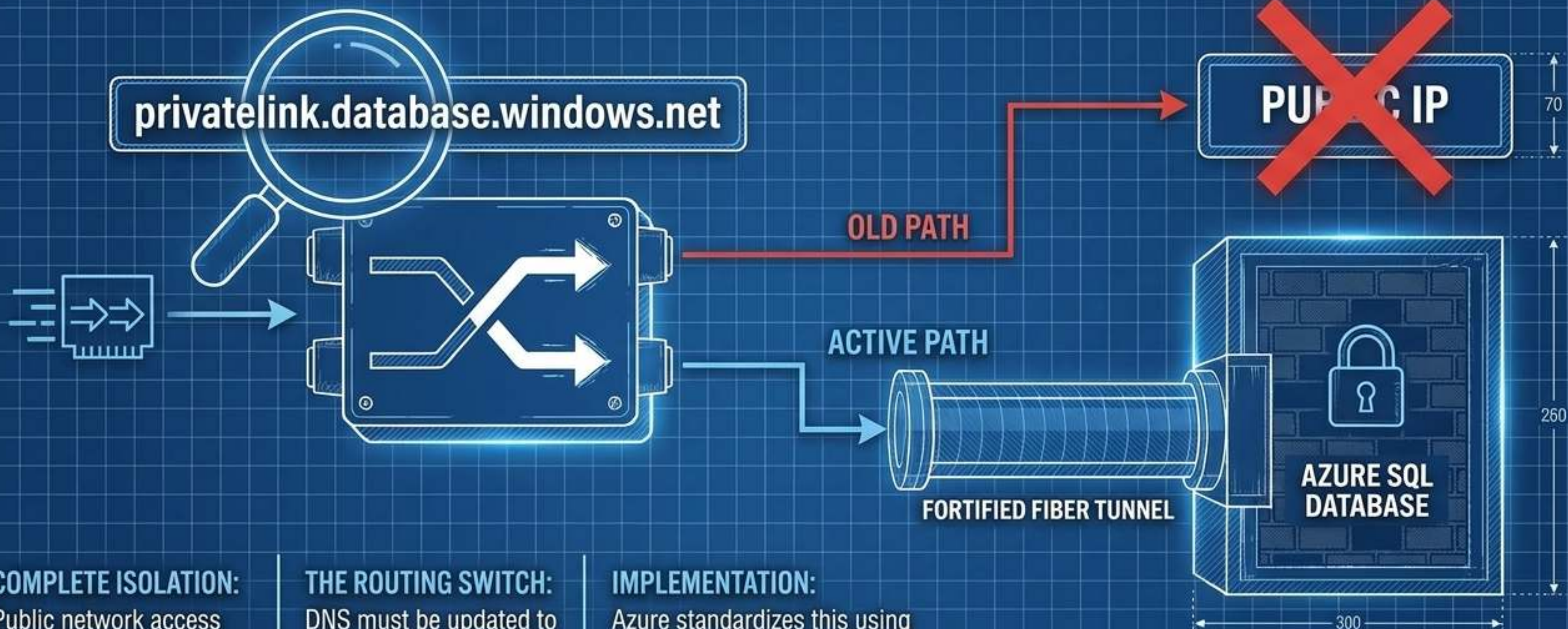
Mapped to a specific resource instance (e.g., your exact SQL database, not all Azure SQL).

THE BENEFIT:

Traffic never touches a public IP. Fundamentally eliminates the risk of data exfiltration to rogue tenants.



THE PRIVATE ENDPOINT REQUIREMENT: DNS



COMPLETE ISOLATION:

Public network access can be 100% disabled on the PaaS resource.

THE ROUTING SWITCH:

DNS must be updated to resolve the PaaS FQDN to the new private IP.

IMPLEMENTATION:

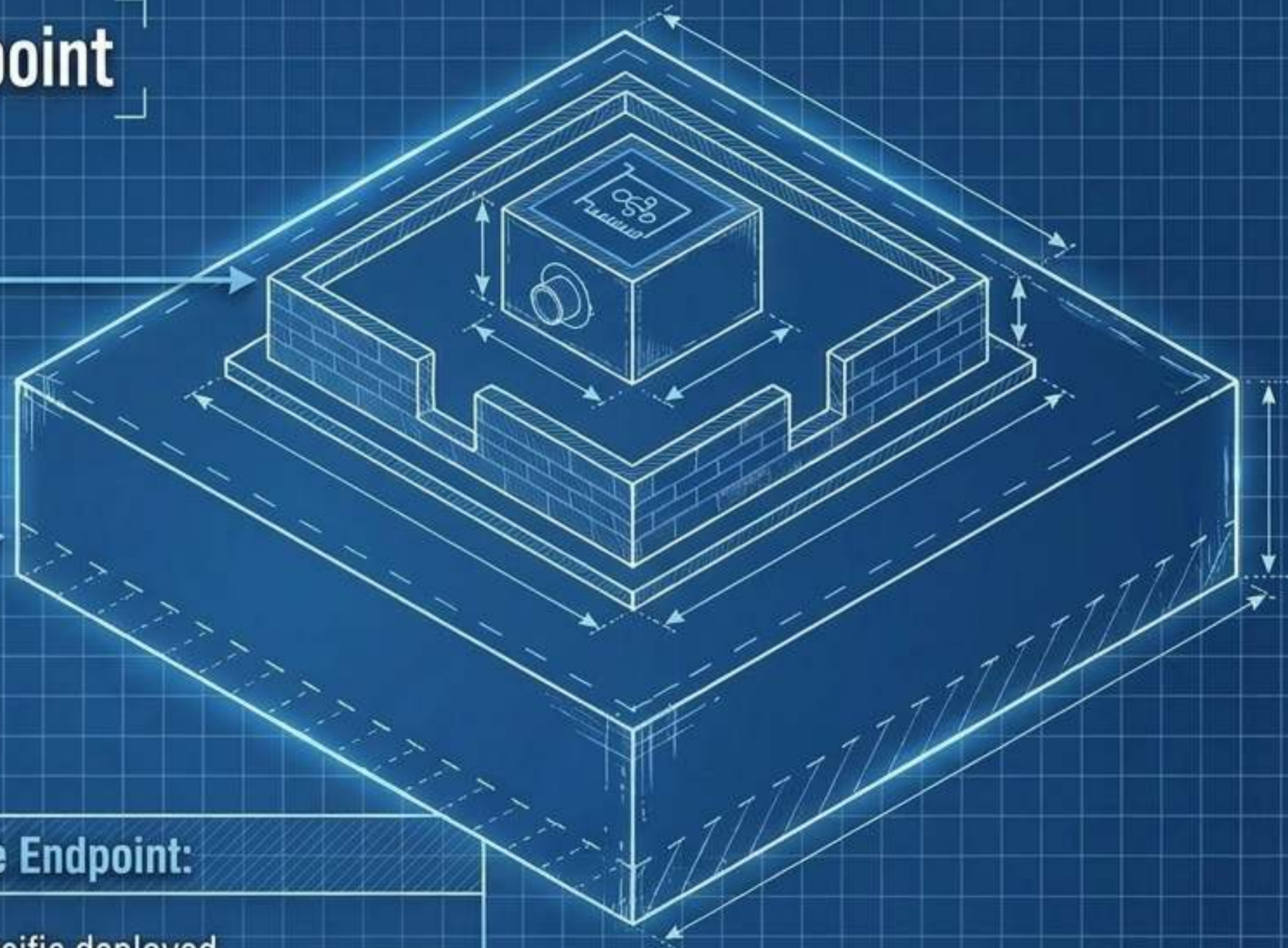
Azure standardizes this using dedicated private DNS zones. Without this DNS update, client traffic will still attempt the public path.

TITLE		PRIVATE ENDPOINT REQUIREMENT: DNS			
SEAL	SEAL	PROJECTED	ROOYFSGTGR	SATE	SSER
		AZURE SECURITY ARCHITECTURE			

Private Link vs. Private Endpoint

Private Endpoint
The Faucet

Azure Private Link
The Plumbing



Azure Private Link:

The underlying platform and technology that enables private connectivity across the Microsoft backbone.



Private Endpoint:

The specific deployed resource (the VNet network interface) that utilizes Private Link to bring a service into your network.



			TITLE				
			PRIVATE ENDPOINT REQUIREMENT: DNS				
SCALE			PROJECTED		ROOYECTOR	DATE	SSSER
SEATE			AZURE SECURITY ARCHITECTURE				

Private Link Service: Exposing Your Own Apps

The Use Case: Securely publish custom or third-party services to other VNets or tenants without public exposure.

The Requirement: Must be deployed behind a Standard Load Balancer.

The Routing Advantage: Uses Source NAT (SNAT) to map traffic. This completely eliminates overlapping IP address conflicts between partner networks, bypassing the primary limitation of VNet peering.



		TITLE			PRIVATE LINK SERVICE: EXPOSING APPS	
SEALB		PROJECTED	PROOFECTOR	STATE	SSSEK	
SEATE		AZURE SECURITY ARCHITECTURE				

The AZ-500 Architectural Matrix

Feature	Public IP Exposure Remains?	Scope	DNS Changes Needed?	Reach Non-Microsoft Services?	Typical Use Case
Service Endpoint	Yes	Entire Subnet	No	No	Routing optimization / Low friction setup
Private Endpoint	No (Can be disabled)	Specific Resource	Yes	Yes (via PL Service)	Full Zero Trust isolation / Data exfiltration prevention

			TITLE			
SCAIB			PROD/ESTED	ROOVECTOR	S0TE	8SSEX
SEATE			A2URE SECURITY ARCHITECTURE			

Architectural Scenario 1: Zero Exposure

Scenario:

You must secure an Azure Storage account. Security policy dictates that you must fully eliminate all public exposure and prevent the resource from being reachable via the internet under any circumstances.

Correct Choice: Private Endpoint

Architectural Justification:

It is the only option that allows you to completely disable the resource's "Public Network Access" flag. Traffic routes exclusively through the VNet NIC, permanently sealing the public doorway.

		TITLE		ARCHITECTURAL SCENARIO 1: ZERO EXPOSURE	
SC01b		PROJECTED	PODVCOTR	NOTE	2SSEX
SC01b		AZURE SECURITY ARCHITECTURE			

Architectural Scenario 2: Low Friction

Scenario:

You need to restrict access to an Azure SQL server so that it only accepts traffic from a specific VNet. However, the network team explicitly forbids the creation or modification of private DNS zones. **without managing private DNS zones**

Correct Choice: Service Endpoint

Architectural Justification:

Service Endpoints apply directly at the subnet level and rely on the Azure fabric to identify traffic. They require zero private DNS configuration to function, perfectly matching the operational constraint.

		TITLE		ARCHITECTURAL SCENARIO 2: LOW FRICTION	
SCM1B		MOD/REVIEWED	POOVVGTOR	SOTE	2SSEX
SC07E		AZURE SECURITY ARCHITECTURE			

Architectural Scenario 3: Partner Integration

Scenario:

You have built an internal billing API hosted on Azure VMs. You must securely expose this API to a partner's VNet. The partner's VNet uses the exact same IP address space as your VNet.

Correct Choice: Private Link Service

Architectural Justification:

Full VNet peering is impossible due to overlapping IP addresses. Private Link Service utilizes a Standard Load Balancer and provider-side publication (via SNAT) to securely bridge the networks without IP routing conflicts.



		TITLE ARCHITECTURAL SCENARIO 3: PARTNER INTEGRATION			
SCALE SCALE		MODIFIED AZURE SECURITY ARCHITECTURE	PROVVISOR	SCALE NOT TO SCALE	SHEET 3 OF 3

AZ-500 Exam Traps

Warning 1: The Service Endpoint Trap

The Trap: Assuming Service Endpoints block public internet access to the resource.

The Truth: The public IP is still resolvable and exposed. The firewall merely drops unapproved traffic. It is not true isolation.

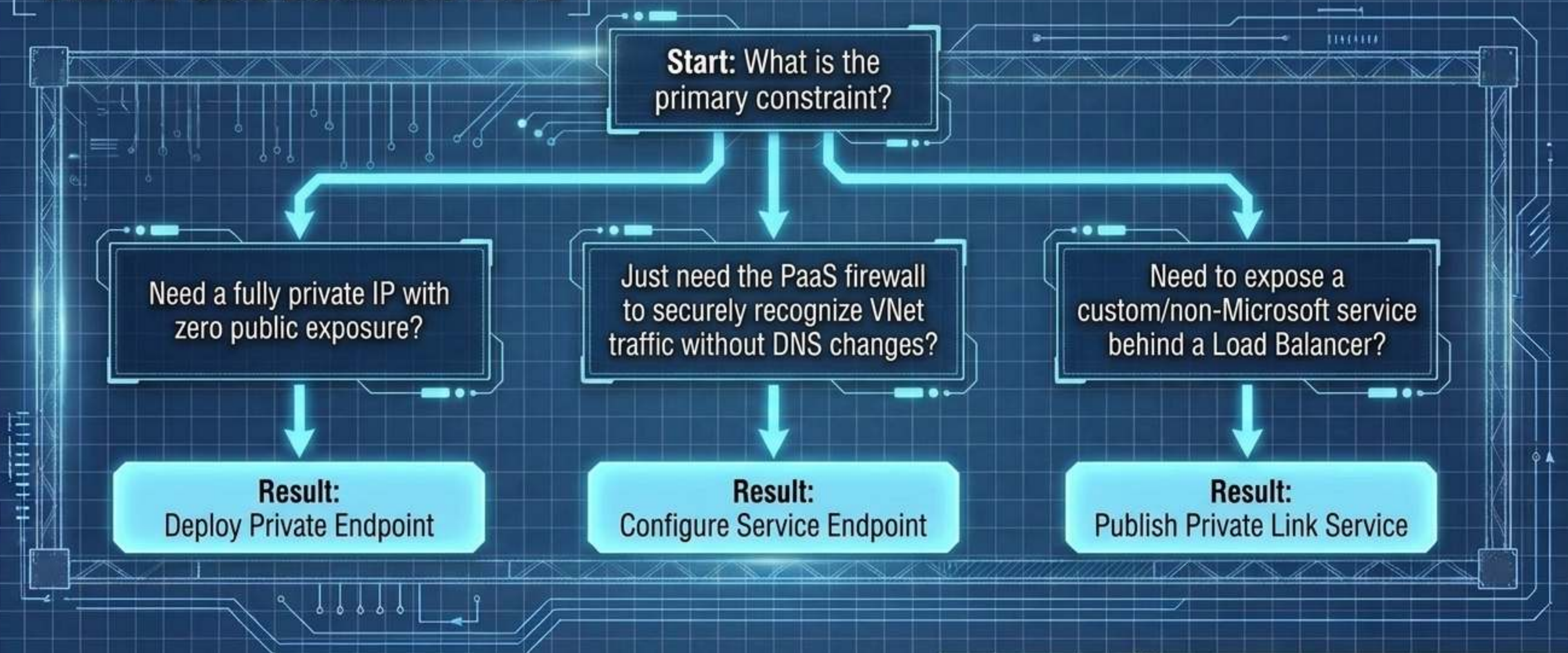
Warning 2: The Nomenclature Trap

The Trap: Treating "Private Link" and "Private Endpoint" as interchangeable answers.

The Truth: Private Link is the underlying network technology. Private Endpoint is the deployable NIC resource you select to secure a specific VNet.

		TITLE		AZURE SECURITY ARCHITECTURE AZ-500 EXAM TRAPS	
SCALE	SCOPE	MOD/EST/D	POW/VTOR	SDAE	36SEX
				NOT TO SCALE	1 of 3
		AZURE SECURITY ARCHITECTURE			

The AZ-500 Decision Flow



TITLE		AZURE SECURITY ARCHITECTURE AZ-500 DECISION FLOW			
SG01B	SG00E	14.0/15.78.0	POOVGTOR	SDAC NOT TO SCALE	SBESX 2 OF 3
		AZURE SECURITY ARCHITECTURE			